

FREDRICK NYANG'AU

Junior Backend Engineer | Python · FastAPI · PostgreSQL · Docker | Fintech and Agritech APIs

Nairobi, Kenya | +254 746 730 585 | nyangaufredrick443@gmail.com | fredricknyangau.vercel.app | github.com/fredricknyangau | linkedin.com/in/fredricknyangau

PROFESSIONAL PROFILE

Junior Backend Engineer with a BSc in Information Technology (Kabarak University, Dec 2025), building production-deployed REST APIs for the East African market. KukuFiti, my flagship agritech project, is live on Render with three production modules and full Swagger documentation. I focus on the backend layer: PostgreSQL schema design, raw SQL via asyncpg, JWT and OAuth2 authentication, and modular FastAPI architecture. My work targets the problems that matter in Kenya: M-Pesa payment flows, ISP billing automation, and farm management systems built for real production conditions. Open to Junior Backend Engineering roles in Nairobi, remote-first teams, and freelance backend contracts in Fintech and Agritech.

TECHNICAL SKILLS

Languages	Python 3.12 (primary), SQL, JavaScript
Frameworks	FastAPI (async-first, high-performance REST APIs)
Databases	PostgreSQL 16, asyncpg, raw SQL, schema design, indexing, query optimization
Auth	JWT, OAuth2 scopes, RBAC, bcrypt, REST API security patterns
Caching	Redis: rate limiting, per-tenant isolation, session management
Infrastructure	Docker, Nginx, Ubuntu 24.04 LTS, systemd, SSH, UFW
DevOps	Git, GitHub, Render deployments, environment-based configuration
API and Testing	Pydantic, pytest, Postman, OpenAPI, Swagger
Integrations	M-Pesa Daraja API (STK push, C2B, webhooks), Africa's Talking SMS
Mobile	Flutter, Dart, Android app development (secondary)

KEY PROJECTS

KukuFiti: Agritech Farm Management API

Live on Render

Python | FastAPI | asyncpg | PostgreSQL 16 | Docker | Render

- Shipped a modular FastAPI backend to Render for Kenyan poultry farmers across 3 production modules (batches, finance, health), delivering a publicly accessible API with full Swagger documentation at /docs and zero manual deployment steps.
- Engineered a PostgreSQL 16 schema using asyncpg and raw SQL with schema-per-module isolation, removing ORM overhead and retaining full query control across all production endpoints.
- Replaced Alembic with version-controlled numbered SQL migrations after diagnosing a production schema mismatch on Render, eliminating an entire class of deployment failures and making schema changes reproducible across environments.

WiFi Billing System: ISP Management API

In Progress

Python | FastAPI | PostgreSQL | M-Pesa Daraja 3.0 | MikroTik RouterOS | Docker

- Architected an ISP billing API with OAuth2 scopes and RBAC, delivering role-separated access for admin, reseller, and customer tiers that mirrors production SaaS authorization patterns.
- Integrated M-Pesa Daraja 3.0 C2B for automated payment processing and webhook handling, building a complete KES payment lifecycle from STK push initiation through callback confirmation.
- Connected directly to MikroTik RouterOS for automated voucher generation and session control, removing the manual ISP provisioning workflow entirely.

Mobile Money Gateway

In Progress

Python | FastAPI | PostgreSQL | Redis | Docker

- Designed a multi-tenant proxy layer for Safaricom Daraja, normalizing inconsistent error codes across tenants and guaranteeing reliable webhook delivery to downstream merchants.
- Implemented exponential backoff retry logic and Redis-based per-tenant rate limiting, protecting downstream services from burst traffic and reducing failed webhook notifications under load.

TraceHub: Lost-and-Found Collaboration Platform

Completed (University Capstone)

Node.js | React | Git | Agile

- Led a 4-member agile squad as Team Lead, managing branch strategies, code reviews, and sprint delivery across the full project lifecycle, resulting in on-time submission of a fully functional platform.
- Designed the search and item-matching engine and resolved critical Git merge conflicts during final deployment, unblocking the team during the highest-pressure phase of delivery.

WORK EXPERIENCE

ICT Intern | [Huduma Centre Nakuru](#)

Jan 2025 – Apr 2025 | Nakuru, Kenya

- Supported 5 government service desks (KRA, NTSA, Civil Registration) handling 50-plus citizens daily on iTax, TIMS, and PMIS, reducing counter resolution time during peak queue hours.
- Operated the Queue Management System for hundreds of daily visitors, improving service triage efficiency and cutting unresolved escalations at the central information desk.
- Digitized pension records and processed birth certificate data entries for the Civil Registration Service, contributing to the ongoing modernization of government records infrastructure.

COMMUNITY AND LEADERSHIP

Founder and Team Lead | [Backend Engineers 2026](#)

Jan 2026 – Present | Nairobi, Kenya

- Founded and leads a 3-person backend engineering study group following a structured 52-week roadmap covering FastAPI, PostgreSQL, system design, and production engineering practices, with a target of employment-ready engineers by Q1 2027.
- Runs weekly code reviews and architecture discussions, with all members maintaining daily Git contribution streaks and progressing through project-based learning milestones.

WRITING AND PUBLICATIONS

- Modular Monolith Backend Systems. Dev.to. Covers architectural decisions for modular FastAPI backends with schema-per-module PostgreSQL isolation and reproducible deployment practices.

EDUCATION

BSc Information Technology | Kabarak University

Sep 2021 – Dec 2025 | Nakuru, Kenya

Second Class Honours (Lower Division) | Certificate No: BScIT 2025/0000070

Relevant coursework: Distributed Systems, Data Structures and Algorithms, Software Engineering, Database Management Systems, Computer Networks, Mobile Applications Programming

REFERENCES

Available on request. Professional: Huduma Kenya, Nakuru Supervisor. Academic: Kabarak University Lecturer.